Subject Description Form

Subject Code	AMA4390
Subject Title	Quantitative Finance and Financial Technology
Credit Value	3
Level	4
Pre-requisite	Derivative Pricing (AMA4325) OR Mathematics for Financial Derivatives (AMA435)
Objectives	The subject will introduce the undergraduate student to the fields of quantitative finance and financial technology.
Intended Learning Outcomes	Upon satisfactory completion of the subject, students should be able to: a) master some basic concepts and products in quantitative finance and financial technology; b) Develop modeling skills in quantitative finance and financial technology; c) Gain a deep understanding of the mathematical methods involved and learn how to apply them in the fields. These methods include ordinary differential equations, partial differential equations, stochastic processes, stochastic calculus, stochastic control, finite difference method, Monte-Carlo simulations, Bayesian analysis, etc.
Subject Synopsis/ Indicative Syllabus	1. Real options; 2. Credit rating and credit spread; 3. Implied volatility, volatility swap, and VIX; 4. Leveraged ETF; 5. Products in crypto-currency markets: automatic market makers, perpetual contracts, and stable coins; 6. Black-Litterman model and robo-advising; 7. Dynamic asset allocation with reinforcement learning; 8. Algorithmic trading.
Teaching/Learning Methodology	The subject will mainly be delivered through lectures and tutorials. The lectures will be conducted to introduce the theoretical background, and practical problems / scenarios will be discussed in the tutorial sessions to illustrate how the theory developed can be applied in practice.

Assessment Methods in Alignment with Intended Learning Outcomes

Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)		
		a	b	c
1. Assignments/Quizzes	30%	√	√	√
2. Midterm Test	20%	√	√	√
3. Examination	50%	√	√	√
Total	100%			

Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:
The subject focuses on knowledge, skill and understanding of Quantitative Finance and Financial Technology and, thus, **Exam-based assessment** is the most appropriate assessment method, including 20% test and 50% examination. Moreover, 30% worth of assignments/quizzes are included as a component of continuous assessment so as to assess students' ability during the semester.

Continuous Assessment comprises of assignments and/or quizzes, and test(s). A written examination is held at the end of the semester.

Student Study Effort Required

Class contact:	
▪ Lecture	26 Hrs.
▪ Tutorial	13 Hrs.
Other student study effort:	
▪ Assignments/Quizzes	35 Hrs.
▪ Self-study	35 Hrs.
Total student study effort	109 Hrs.

Reading List and References

References:

Alvaro Cartea, Sebastian Jaimungal, Jose Penalva	Algorithmic and High-Frequency Trading	Cambridge 2015

	Matthew F. Dixon, Igor Halperin, Paul Bilokon	Machine Learning in Finance	Springer 2020
	John Hull	Options, Futures, and Other Derivatives, 11^{th} edition	Pearson/Prentice Hall 2021
	Steven Shreve	Stochastic Calculus for Finance I: The Binomial Asset Pricing Model	Springer-Verlag 2008
	Steven Shreve	Stochastic Calculus for Finance II: Continuous-Time Models	Springer-Verlag 2010
	Paul Wilmott	Paul Wilmott on Quantitative Finance	Wiley 2006